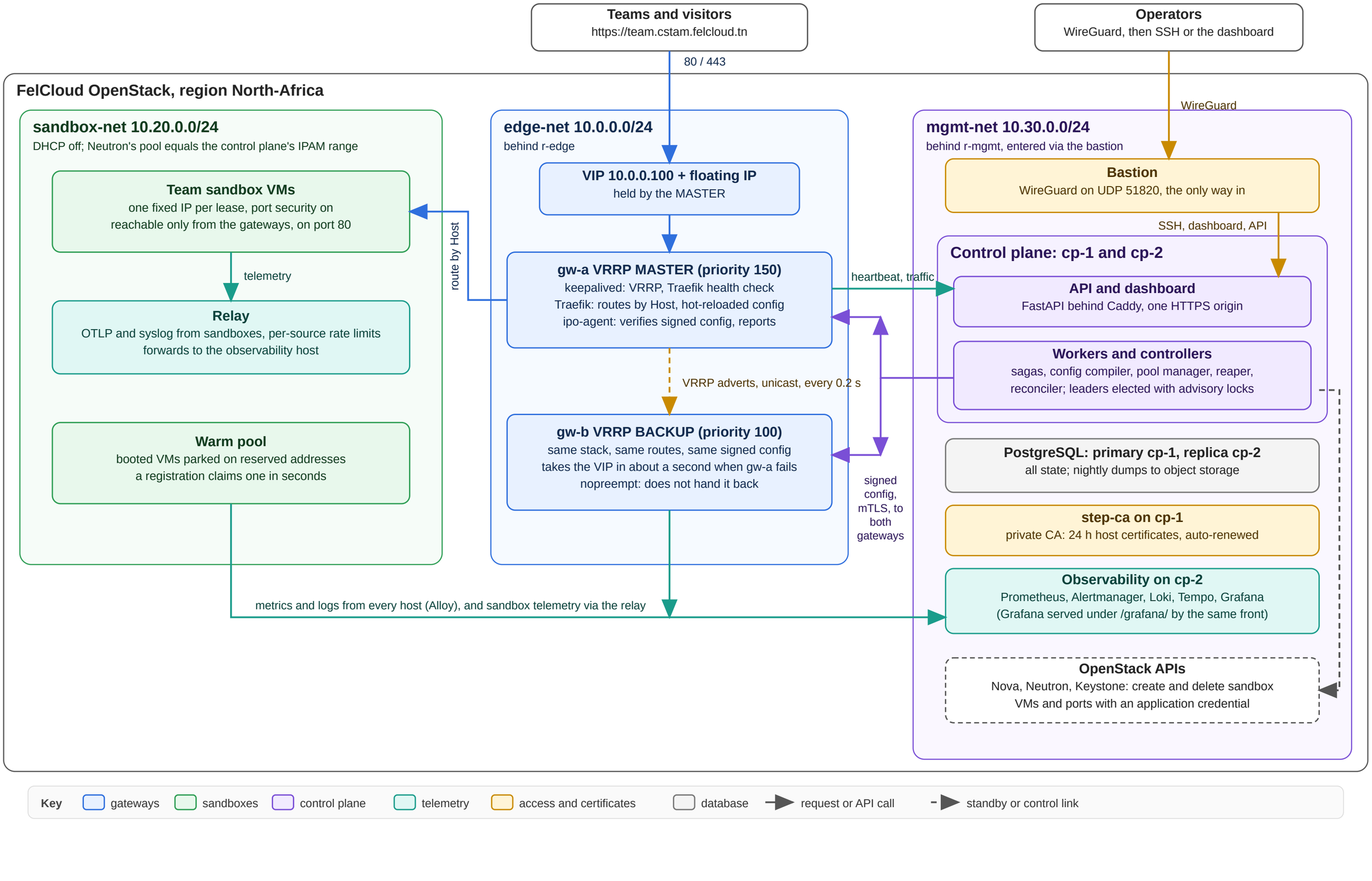
# System architecture

The platform on FelCloud OpenStack: one floating IP, a VRRP gateway pair, a control plane, isolated sandboxes.



# OpenStack high availability

One floating IP and one VIP port; two gateways that can each hold the VIP.

## Normal

gw-a is the MASTER, gw-b is hot standby

**Visitors**
HTTP(S) to the platform's public name

**Floating IP**
attached to the vip-edge port, not to a VM

**vip-edge port 10.0.0.100**
no device: reserved, so nothing else can take it

**gw-a MASTER**
holds the VIP
Traefik serving, adverts every 0.2 s
edge port: allowed_address_pairs = VIP

compute host A

**gw-b BACKUP**
listens for adverts
Traefik loaded, routes identical
edge port: allowed_address_pairs = VIP

compute host B, if the cloud has one

dashed: VRRP adverts every 0.2 s

## After losing gw-a's VM

gw-b takes the VIP; clients reconnect to the same address

**Visitors**
HTTP(S) to the platform's public name

**Floating IP**
attached to the vip-edge port, not to a VM

**vip-edge port 10.0.0.100**
no device: reserved, so nothing else can take it

**gw-a LOST**
VM gone: processes dead,
NICs dark, adverts stop
edge port: allowed_address_pairs = VIP

compute host A

**gw-b MASTER**
holds the VIP
Traefik serving, adverts every 0.2 s
edge port: allowed_address_pairs = VIP

compute host B, if the cloud has one

gw-b takes the VIP in about 0.6 s; measured worst gap 0.52 to 0.66 s at 50 requests/s

## What the handover needs from OpenStack

**Allowed address pair**
Both gateways' edge ports list the VIP, so port security lets them send and receive it

**A device-less VIP port**
Holds 10.0.0.100 and the floating IP, so the address survives either VM

**Unicast VRRP**
Multicast is often filtered, so adverts go unicast; the group admits VRRP

**FAULT releases at once**
Check script with no weight: two failed checks = FAULT = priority-0 advert, no wait

**nopreempt**
The old MASTER returns as BACKUP, so a recovery is not a second outage

## What is redundant, and what is not

green: redundant, amber: partly or manual, red: single instance

**Gateways**
VRRP pair, anti-affinity
(soft: see the HA document)
loss: about 0.6 s

**Control plane**
two active replicas, leaders by advisory lock; gateways serve without it

**PostgreSQL**
streaming replica on cp-2;
promote_replica.yml is manual;
writes pause meanwhile

**ipo-agent**
last-known-good config kept;
restart resumes it;
systemd restarts it

**Sandbox VM**
one team's outage only;
reconciler repairs drift;
warm pool refills

**CA, observability**
single instance each:
24 h certificates give a window; not redundant

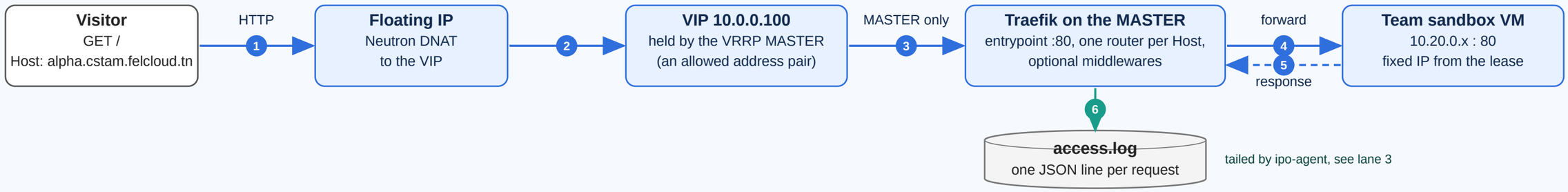
Key — MASTER or redundant · BACKUP or partly redundant · lost or single instance · traffic path · needed from OpenStack · traffic · VRRP adverts

# Gateway data flow

Requests, configuration, and the signals and telemetry that flow back. gw-b runs the same stack as the standby.

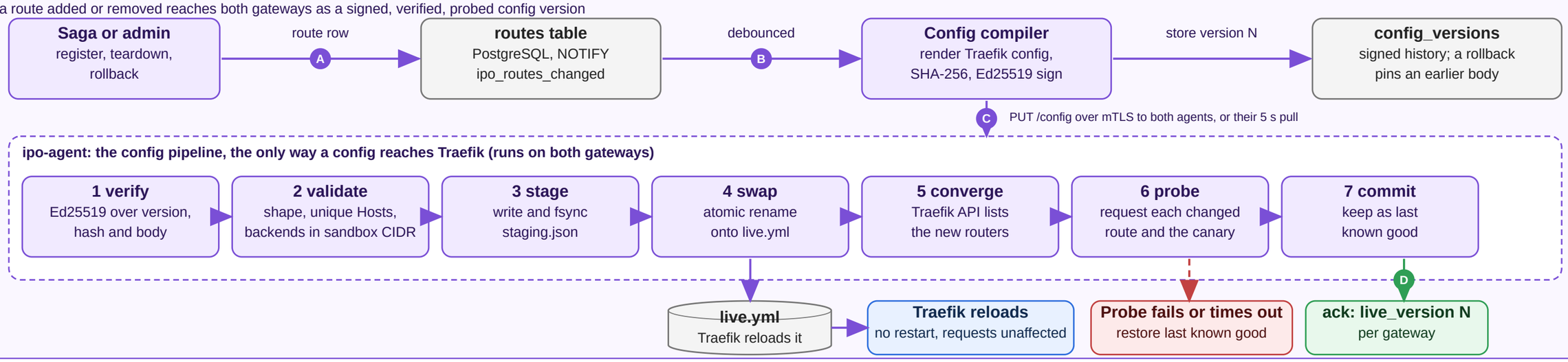
## 1 Request path

a visitor reaching a team's sandbox, through whichever gateway holds the VIP




## 2 Configuration path

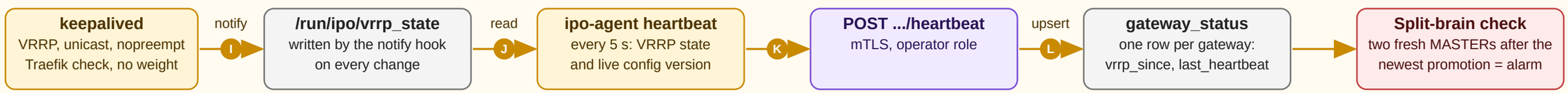
a route added or removed reaches both gateways as a signed, verified, probed config version



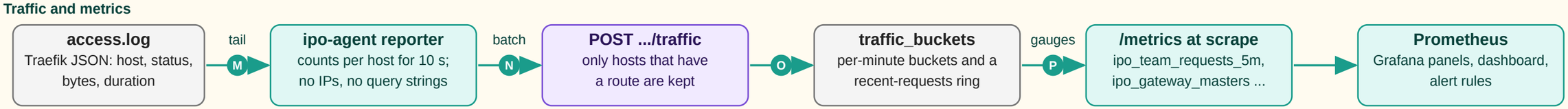

## 3 Signals and telemetry

what the gateways report back, and what the control plane does with it

### VRRP state and liveness




### Traffic and metrics




Key: request path · configuration · state and liveness · telemetry · stored data · failure outcome · success · data flow · response or failure · 1 step order